Day 10 Evidence Workbooklet

Week 2 Decision Evidence

Engineering practice to branch map to case table to test suite to Week 3 bridge

Student copy. Guided workbooklet, not the mastery quiz. Print format: duplex, left-stapled packet.

Name		**Section**		**Date**	
Score	______/50		Mastery check	secure / developing / needs support	

Concrete engineering situation

Day 10 closes the Week 2 sensor-kit checkout scenario. The team needs a route map, a case table, tests, and a short decision note before using the checkout process on many kits. The focus is consolidation: branch routing, first-true order, testing, debugging, and explaining why repeated cases will need loops next week.

Engineering task	Judge whether a sensor-kit checkout route is ready to use on real checkout records and what evidence should be kept.
Computational move	Synthesize branch maps, case traces, tests, and repair logic into a bounded decision-evidence packet.
Python focus	`if`/`elif`/`else`, first-true route, test suite, debug repair, closeout note, and bridge to repeated cases.
Evidence to keep	branch map, case trace, first-true reason, boundary test, regression test, bug cause, repair, and Week 3 bridge note.

Day 10 working rules

1. Use the route map before judging a case.
2. Stop at the first true rule.
3. Include tests for risks, boundaries, and clean release.
4. A wrong release is usually a rule-order or missing-condition problem.
5. Repeated cases are the reason Week 3 introduces loops, not a reason to start loops today.

1 Read the Week 2 release map

recognize predict

Day 10 closes the sensor-kit checkout scenario. The goal is not new syntax. The goal is to use branch routing, ordered validation, expected-vs-actual evidence, and tests in one bounded decision surface.

```
if damage_note != "":
    route = "repair bench"
elif charge_percent < 80:
    route = "charge station"
elif calibration_label != "current":
    route = "calibration bench"
elif borrower_initials == "":
    route = "borrower log needed"
else:
    route = "release kit"
```

Pts.	Prompt	My evidence
1	What route has highest priority?	
1	What route happens when the kit is low on charge and not damaged?	
1	What route happens when every hold condition is false?	
1	What Week 3 problem will appear if there are many kits to check?	

2 Complete a decision case table

trace explain

Trace each kit through the release map. Stop at the first true rule.

Pts.	Kit	Charge	Calibration	Other evidence	Route and reason
2	A	92	current	no damage, initials present	
2	B	92	current	cracked case, initials present	
2	C	78	expired	no damage, initials present	
2	D	88	current	no damage, blank initials	

3 Plan a small test suite

test debug

Choose tests that make the release map trustworthy before the desk uses it on many kits.

Pts.	Prompt	My test
2	One test for the highest-priority route.	
2	One boundary test for charge exactly at `80`.	
2	One test for an expired calibration label.	
2	One test that should release the kit.	

Test-suite warning

A test suite that only checks release-ready kits can pass even when every hold route is broken. Include risks, boundaries, and a clean release case.

4 Debug and repair a wrong release

debug implement

A damaged kit was released. Read the wrong version and identify the repair.

```
if charge_percent >= 80 and calibration_label == "current" and borrower_initials != "":
    route = "release kit"
elif damage_note != "":
    route = "repair bench"
elif charge_percent < 80:
    route = "charge station"
else:
    route = "hold for review"
```

Pts.	Prompt	My response
2	What route did the wrong version choose for a damaged but otherwise ready kit?	
2	What route should it choose?	
2	What is the smallest rule-order repair?	
2	Which test from Section 3 would catch this bug?	

5 Write the Week 2 decision evidence note

transfer explain

Field	My Week 2 evidence
Branch map summary	
Most important first-true rule	
Most important test to keep	
Likely bug to watch for	
Decision about using the route on real checkout records	

6 Transfer and support route

transfer explain

Pts.	Week 2 closeout note
5	Write a closeout note that names the route decision, cites branch-order evidence and one test, and explains why Week 3 will need repeated-case tracing.
2	Circle the support route you would use next: branch map / first-true trace / test suite / bug cause / repair / repeated cases. Name one next action: ________________

Bridge to Week 3

Week 2 decided routes one case at a time. Week 3 asks what happens when the same route logic must be traced across repeated cases. That is where loops become useful.